Description about conservative horizontal momentum diffusion scheme and tensorial stress option in LFRic

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1 Equation of diffusion scheme

In LFRic's Smagorinsky scheme and blending boundary layer scheme [1], subgrid turbulent mixing of momentum is calculated by the following equation:

$$\frac{\partial u_i}{\partial t} = -\frac{1}{\rho}\frac{\partial F^{u_i,x_j}}{\partial x_j} \qquad \left(F^{u_i,x_j} = -\rho\nu\frac{\partial u_i}{\partial x_j}\right), \tag{1}$$

where ρ and ν are density and mixing coefficient, respectively. u_i and x_i are 3-dimensional velocity and coordinate following Einstein's notation rule. Imcompressive formulation is adopted, and density is ignored in the horizontal diffusion, so each F^{u_i,x_j} is represented as follows:

$$F^{ux} = -\nu\frac{\partial u}{\partial x}, \quad F^{uy} = -\nu\frac{\partial u}{\partial y}, \quad F^{uz} = -\nu\frac{\partial u}{\partial z},$$

$$F^{vx} = -\nu\frac{\partial v}{\partial x}, \quad F^{vy} = -\nu\frac{\partial v}{\partial y}, \quad F^{vz} = -\nu\frac{\partial v}{\partial z},$$

$$F^{wx} = -\nu\frac{\partial w}{\partial x}, \quad F^{wy} = -\nu\frac{\partial w}{\partial y}, \quad F^{wz} = -\nu\frac{\partial w}{\partial z},$$

where x, y, z, u, v and w are 3-dimensional coordinate and wind components, respectively. Since, red colored terms are calculated not in the diffusion scheme but in the boundary layer scheme, we ignore them in the following explanation. The equations for u, v, w are as follows:

$$\frac{\partial u}{\partial t} = -\frac{\partial F^{ux}}{\partial x} - \frac{\partial F^{uy}}{\partial y}, \tag{2}$$

$$\frac{\partial v}{\partial t} = -\frac{\partial F^{vx}}{\partial x} - \frac{\partial F^{vy}}{\partial y}, \tag{3}$$

$$\frac{\partial w}{\partial t} = -\frac{\partial F^{wx}}{\partial x} - \frac{\partial F^{wy}}{\partial y}. \tag{4}$$

2 Conservative horizontal diffusion

Here, we will explain how to calculate (2), (3) and (4). In the following, as shown in Figure1, locations that are half-grid shifted relative to grid numbers i, j and k will be represented as $i + \frac{1}{2}, j + \frac{1}{2}$ and $k + \frac{1}{2}$, respectively.

In LFRic, u, v and w are placed on the face of each cell as shown in Figure2.

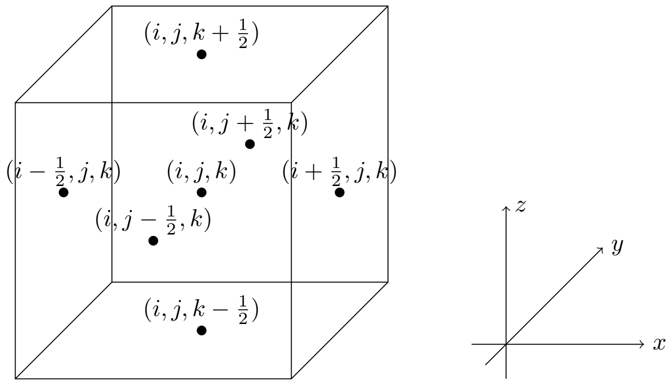


Figure1: Notation of grid number. Cube indecates (i, j, k)th cell.

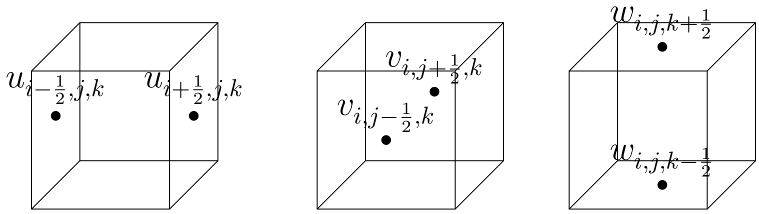


Figure2: Definition point of each velocity on the face of (i, j, k)th cell.

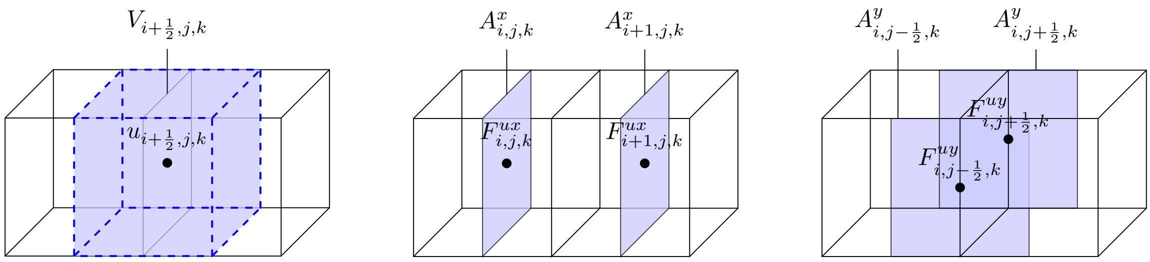
In the calculation of (2) for $u_{i+\frac{1}{2},j,k}$, we assume a volume $V_{i+\frac{1}{2},j,k}$ surrounding $u_{i+\frac{1}{2},j,k}$. ν is linearly interpolated to the face of $V_{i+\frac{1}{2},j,k}$, and wind shears are also evaluated at the same locations. F^{ux} and F^{uy} are calculated on the face of $V_{i+\frac{1}{2},j,k}$ by multiplying them as depected in Figure3. Increment of $u_{i+\frac{1}{2},j,k}$ is calculated as:

$$\Delta u_{i+\frac{1}{2},j,k} = -\frac{\left(F^{ux}_{i+1,j,k}A^x_{i+1,j,k} - F^{ux}_{i,j,k}A^x_{i,j,k}\right) + \left(F^{uy}_{i,j+\frac{1}{2},k}A^y_{i,j+\frac{1}{2},k} - F^{uy}_{i,j-\frac{1}{2},k}A^y_{i,j-\frac{1}{2},k}\right)}{V_{i+\frac{1}{2},j,k}}\Delta t, \tag{5}$$

where A^x and A^y are the area of faces on which F^{ux} and F^{uy} are placed, respectively. Please see Figure3 for placement of each term in (5). Since adjacent cells share the same $F^{ux}A^x$ and $F^{uy}A^y$ at their boundary, the increment of total volume-weighed u is cancelled out by summing up (5) and ignoring the effect of lateral boundary condition:

$$\Delta\left(\sum_{i,j} V_{i+\frac{1}{2},j,k}u_{i+\frac{1}{2},j,k}\right) = \sum_{i,j} V_{i+\frac{1}{2},j,k}\Delta u_{i+\frac{1}{2},j,k} = 0, \tag{6}$$

which means (5) conserves volume-weighed total u.



(a) Blue cube indicates $V_{i+\frac{1}{2},j,k}$, and black dot indicates where $u_{i+\frac{1}{2},j,k}$ is located.

(b) Blue area indicates A^x, and black dot indicates where F^{ux} is calculated.

(c) Blue area indicates A^y, and black dot indicates where F^{uy} is calculated.

Figure3: Placement of each terms appered in (5).

In the calculation of $\Delta v_{i,j+\frac{1}{2},k}$ and $\Delta w_{i,j,k+\frac{1}{2}}$, we assume volumes $V_{i,j+\frac{1}{2},k}$ and $V_{i,j,k+\frac{1}{2}}$, respectively, and calculate $F^{vx}, F^{vy}, F^{wx}, F^{wy}, A^x$ and A^y on their faces. Increments of $v_{i,j+\frac{1}{2},k}$ and $w_{i,j,k+\frac{1}{2}}$ are calculated in the same manner as (5):

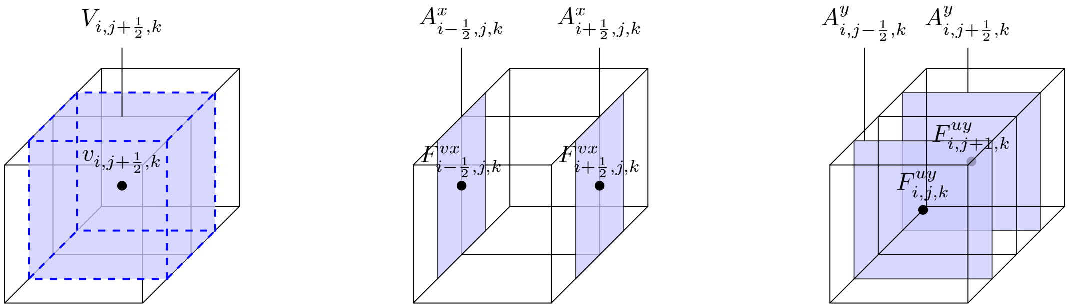
$$\Delta v_{i,j+\frac{1}{2},k} = -\frac{\left(F^{vx}_{i+\frac{1}{2},j,k} A^x_{i+\frac{1}{2},j,k} - F^{vx}_{i-\frac{1}{2},j,k} A^x_{i-\frac{2}{2},j,k}\right) + \left(F^{vy}_{i,j+1,k} A^y_{i,j+1,k} - F^{vy}_{i,j,k} A^y_{i,j,k}\right)}{V_{i,j+\frac{1}{2},k}} \Delta t, \tag{7}$$

$$\Delta w_{i,j,k+\frac{1}{2}} = -\frac{\left(F^{wx}_{i+\frac{1}{2},j,k+\frac{1}{2}} A^x_{i+\frac{1}{2},j,k+\frac{1}{2}} - F^{wx}_{i-\frac{1}{2},j,k+\frac{1}{2}} A^x_{i-\frac{2}{2},j,k+\frac{1}{2}}\right) + \left(F^{wy}_{i,j+\frac{1}{2},k+\frac{1}{2}} A^y_{i,j+\frac{1}{2},k+\frac{1}{2}} - F^{wy}_{i,j-\frac{1}{2},k+\frac{1}{2}} A^y_{i,j-\frac{1}{2},k+\frac{1}{2}}\right)}{V_{i,j,k+\frac{1}{2}}} \Delta t, \tag{8}$$

Please see Figure4 and Figure5 for placement of each term in (7) and (8), respectively.. Similarly, volume-weighed total v and w are conserved through the calculation.

$$\Delta\left(\sum_{i,j} V_{i,j+\frac{1}{2},k} v_{i,j+\frac{1}{2},k}\right) = \sum_{i,j} V_{i,j+\frac{1}{2},k} \Delta v_{i,j+\frac{1}{2},k} = 0 \tag{9}$$

$$\Delta\left(\sum_{i,j} V_{i,j,k+\frac{1}{2}} w_{i,j,k+\frac{1}{2}}\right) = \sum_{i,j} V_{i,j,k+\frac{1}{2}} \Delta w_{i,j,k+\frac{1}{2}} = 0 \tag{10}$$

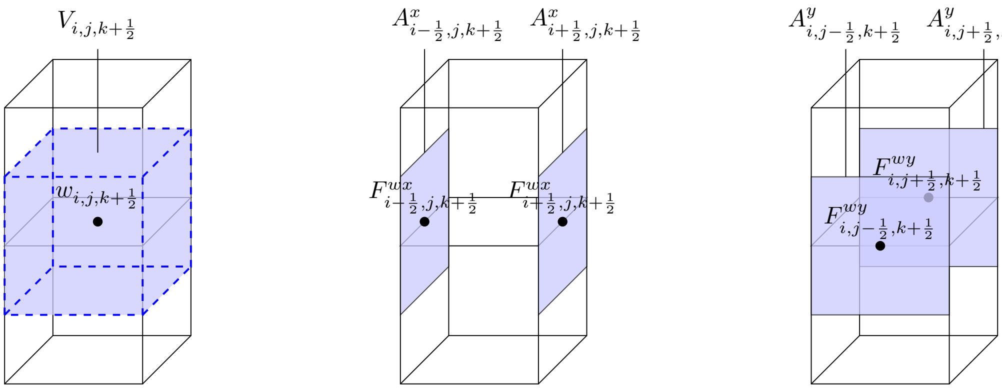


(a) Blue cube indicates $V_{i,j+\frac{1}{2},k}$, and black dot indicates where $v_{i,j+\frac{1}{2},k}$ is located.

(b) Blue area indicates A^x, and black dot indicates where F^{vx} is calculated.

(c) Blue area indicates A^y, and black dot indicates where F^{uy} is calculated.

Figure4: Placement of each terms appered in (7).



(a) Blue cube indicates $V_{i,j,k+\frac{1}{2}}$, and black dot indicates where $w_{i,j,k+\frac{1}{2}}$ is located.

(b) Blue area indicates A^x, and black dot indicates where F^{wx} is calculated.

(c) Blue area indicates A^y, and black dot indicates where F^{uy} is calculated.

Figure5: Placement of each terms appered in (8).

3 Tensorial momentum diffusion option

Here, we will explain the equation of tensorial diffusion equation and how to calculate it. When tensorial stress is used, (1) will be:

$$\frac{\partial u_i}{\partial t} = -\frac{1}{\rho}\frac{\partial F^{u_i,x_j}}{\partial x_j} \qquad \left(F^{u_i,x_j} = -\rho\nu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)\right), \tag{11}$$

where blue colored terms are newly added in the tensorial diffusion. Adopting imcompressive formulation, each $F^{u_i x_j}$ is represented as:

$$F^{ux} = -2\nu\frac{\partial u}{\partial x}, \quad F^{uy} = -\nu\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right), \quad F^{uz} = -\nu\left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z}\right),$$

$$F^{vx} = -\nu\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right), \quad F^{vy} = -2\nu\frac{\partial v}{\partial y}, \quad F^{vz} = -\nu\left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}\right),$$

$$F^{wx} = -\nu\left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z}\right), \quad F^{wy} = -\nu\left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}\right), \quad F^{uz} = -2\nu\frac{\partial w}{\partial z}.$$

Ignoring red colored terms, which are calculated not in the diffusion scheme but in the boundary layer scheme, the equations for u, v and w will be:

$$\frac{\partial u}{\partial t} = -\frac{\partial F^{ux}}{\partial x} - \frac{\partial F^{uy}}{\partial y} - \frac{\partial F^{uz*}}{\partial z} \qquad \left(F^{uz*} = -\nu\frac{\partial w}{\partial x}\right), \tag{12}$$

$$\frac{\partial v}{\partial t} = -\frac{\partial F^{vx}}{\partial x} - \frac{\partial F^{vy}}{\partial y} - \frac{\partial F^{vz*}}{\partial z} \qquad \left(F^{vz*} = -\nu\frac{\partial w}{\partial y}\right), \tag{13}$$

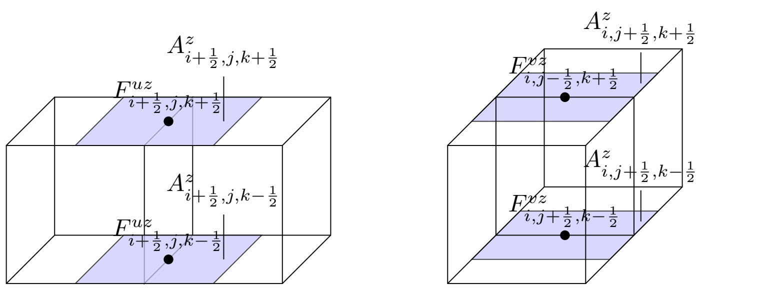
$$\frac{\partial w}{\partial t} = -\frac{\partial F^{wx}}{\partial x} - \frac{\partial F^{wy}}{\partial y}. \tag{14}$$

We can calculate the first and the second terms on the right hand side of (12), (13) and (14) in the same way as (5), (7) and (8), respectively, except for adding blue colored terms in $F^{u_i x_j}$. In the discritization of the third terms in (12), (13), we adopt a similar conservative way as follows:

$$-\left.\frac{\partial F^{uz*}}{\partial z}\right|_{i+\frac{1}{2},j,k} = -\frac{F^{uz*}_{i+\frac{1}{2},j,k+\frac{1}{2}} A^z_{i+\frac{1}{2},j,k+\frac{1}{2}} - F^{uz*}_{i+\frac{1}{2},j,k-\frac{1}{2}} A^z_{i+\frac{1}{2},j,k-\frac{1}{2}}}{V_{i+\frac{1}{2},j,k}} \tag{15}$$

$$-\left.\frac{\partial F^{vz*}}{\partial z}\right|_{i,j+\frac{1}{2},k} = -\frac{F^{vz*}_{i,j+\frac{1}{2},k+\frac{1}{2}} A^z_{i,j+\frac{1}{2},k+\frac{1}{2}} - F^{vz*}_{i,j+\frac{1}{2},k-\frac{1}{2}} A^z_{i,j+\frac{1}{2},k-\frac{1}{2}}}{V_{i,j+\frac{1}{2},k}} \tag{16}$$

Please see Figure6 for placement of each term in (15) and (16).



(a) Blue area indicates A^z, and black dot indicates where F^{uz} is calculated.

(b) Blue area indicates A^z, and black dot indicates where F^{vz} is calculated.

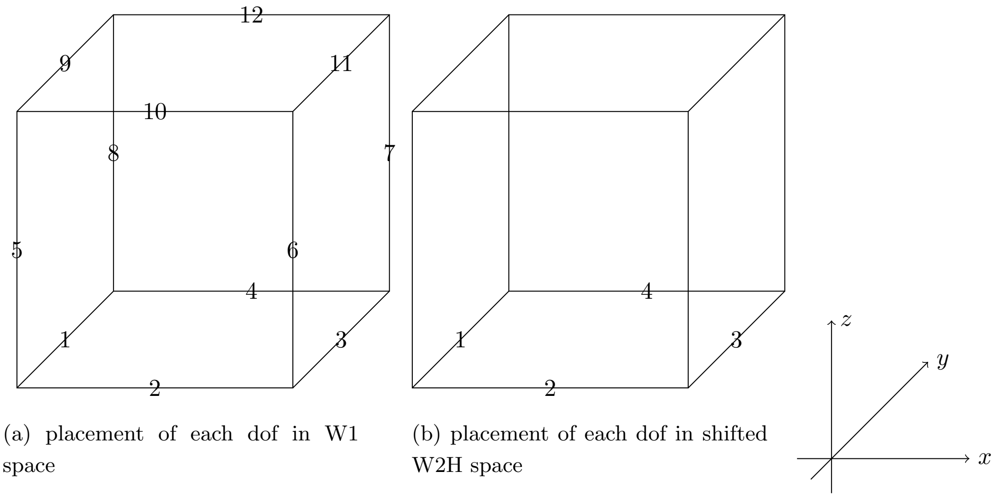
Figure6: Placement of each terms appered in (15) and (16).

4 Usage of function space

In the LFRic's source code, we use $W3$ function spaces for variables placed on the center of the cell, while we use $W1$ or shifted $W2H$ function space for variables placed on the edges of the cell. Each degree of freedom (dof) in $W1$ and shifted $W2H$ function space is placed as shown in Figure7. Function spaces used for each $F^{u_i x_j}$ is listed in table1. As shown in table1, we sometimes treat 2 variables together in a single variable (e.g. we treat F^{uy} and F^{uz*} in a single variable uflux_w1).

Notation in this document	Placement	Variable name in the source code	function space
F^{ux}	(i,j,k)	uflux_w3	$W3$
F^{uy}	$(i+\frac{1}{2}, j+\frac{1}{2}, k)$	uflux_w1	$W1$
F^{uz*}	$(i+\frac{1}{2}, j, k+\frac{1}{2})$		
F^{vy}	(i,j,k)	vflux_w3	$W3$
F^{vx}	$(i+\frac{1}{2}, j+\frac{1}{2}, k)$	vflux_w1	$W1$
F^{vz*}	$(i, j+\frac{1}{2}, k+\frac{1}{2})$		
F^{wx}	$(i+\frac{1}{2}, j, k+\frac{1}{2})$	wflux_sh_w2h	shifted $W2H$
F^{wy}	$(i, j+\frac{1}{2}, k+\frac{1}{2})$		

Table1: Placement and function space for each variable.



(a) placement of each dof in W1 space

(b) placement of each dof in shifted W2H space

Figure7: placement of dof

References

[1] I. A. Boutle, J. E.Eyre, and A. Lock. Seamless stratocumulus simulation across the turbulent gray zone. *Monthly Weather Review*, 142(4):1655–1668, 2014.